

Cheng Zhang

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Education

University of California, Irvine

PH.D. IN COMPUTER SCIENCE

Supervisor: Prof. Shuang Zhao

Irvine, CA

September 2017 – July 2022

Columbia University in the City of New York

M.S. IN COMPUTER SCIENCE

GPA: 3.97/4.0

New York, NY

September 2015 – February 2017

Beijing University of Technology

B.E. IN ELECTRONICS INFORMATION ENGINEERING

Beijing, China

September 2011 – June 2015

- GPA: 3.74/4.0

- Exchange student at **University of Illinois at Chicago** (senior year)

Publications

LSRM: HIGH-FIDELITY OBJECT-CENTRIC RECONSTRUCTION VIA SCALED CONTEXT WINDOWS

Zhengqin Li, **Cheng Zhang**, Jakob Engel, Zhao Dong

European Conference on Computer Vision (ECCV), 2026

ART: ARTICULATED RECONSTRUCTION TRANSFORMER

Zizhang Li, **Cheng Zhang**, Zhengqin Li, Henry Howard-Jenkins, Zhaoyang Lv, Chen Geng, Jiajun Wu, Richard Newcombe, Jakob Engel, Zhao Dong

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2026

IMAGE-SPACE ADAPTIVE SAMPLING FOR FAST INVERSE RENDERING

Kai Yan, **Cheng Zhang**, Sébastien Speierer, Guangyan Cai, Yufeng Zhu, Zhao Dong, Shuang Zhao

ACM Transactions on Graphics (SIGGRAPH), 2025

LIRM: LARGE INVERSE RENDERING MODEL FOR PROGRESSIVE RECONSTRUCTION OF SHAPE, MATERIALS AND VIEW-DEPENDENT RADIANCE FIELDS

Zhengqin Li, Dilin Wang, Ka Chen, Zhaoyang Lv, Thu Nguyen-Phuoc, Milim Lee, Jia-Bin Huang, Lei Xiao, **Cheng Zhang**, Yufeng Zhu, Carl S Marshall, Yuheng Ren, Richard Newcombe, Zhao Dong

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025

DIGITAL TWIN CATALOG: A LARGE-SCALE PHOTOREALISTIC 3D OBJECT DIGITAL TWIN DATASET

Zhao Dong, Ka Chen, Zhaoyang Lv, Hong-Xing Yu, Yunzhi Zhang, **Cheng Zhang**, Yufeng Zhu, Stephen Tian, Zhengqin Li, Geordie Moffatt, Sean Christofferson, James Fort, Xiqing Pan, Mingfei Yan, Jiajun Wu, Carl Yuheng Ren, Richard Newcombe

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025

ESTIMATING UNCERTAINTY IN APPEARANCE ACQUISITION

Zhiqian Zhou, **Cheng Zhang**, Zhao Dong, Carl Marshall, Shuang Zhao

Eurographics Symposium on Rendering (EGSR), 2024

TEXTUREDREAMER: IMAGE-GUIDED TEXTURE SYNTHESIS THROUGH GEOMETRY-AWARE DIFFUSION

Yu-Ying Yeh, Jia-Bin Huang, Changil Kim, Lei Xiao, Thu Nguyen-Phuoc, Numair Khan, **Cheng Zhang**, Manmohan Chandraker, Carl S Marshall, Zhao Dong, Zhengqin Li

Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024

NEURAL-PBIR RECONSTRUCTION OF SHAPE, MATERIAL, AND ILLUMINATION

Cheng Sun, Guangyan Cai, Zhengqin Li, Kai Yan, **Cheng Zhang**, Carl Marshall, Jia-Bin Huang, Shuang Zhao, Zhao Dong

International Conference on Computer Vision (ICCV), 2023

SAM-RL: SENSING-AWARE MODEL-BASED REINFORCEMENT LEARNING VIA DIFFERENTIABLE PHYSICS-BASED SIMULATION AND RENDERING

Jun Lv, Yunhai Feng, **Cheng Zhang**, Shuang Zhao, Lin Shao, Cewu Lu

Robotics: Science and Systems (Best System Paper Award Finalist), 2023

EFFICIENT PATH-SPACE DIFFERENTIABLE VOLUME RENDERING WITH RESPECT TO SHAPES

Zihan Yu, **Cheng Zhang**, Olivier Maury, Christophe Hery, Zhao Dong, Shuang Zhao

Computer Graphics Forum (Eurographics Symposium on Rendering), 2023

EFFICIENT DIFFERENTIATION OF PIXEL RECONSTRUCTION FILTERS FOR PATH-SPACE DIFFERENTIABLE RENDERING

Zihan Yu, **Cheng Zhang**, Derek Nowrouzezahrai, Zhao Dong, Shuang Zhao

ACM Transactions on Graphics (*SIGGRAPH Asia 2022*), 41(6)

ANTITHETIC SAMPLING FOR MONTE CARLO DIFFERENTIABLE RENDERING

Cheng Zhang, Zhao Dong, Michael Doggett, Shuang Zhao

ACM Transactions on Graphics (*SIGGRAPH 2021*), 40(4)

PATH-SPACE DIFFERENTIABLE RENDERING OF PARTICIPATING MEDIA

Cheng Zhang^{*}, Zihan Yu^{*}, Shuang Zhao (^{*}equal contribution)

ACM Transactions on Graphics (*SIGGRAPH 2021*), 40(4)

PATH-SPACE DIFFERENTIABLE RENDERING

Cheng Zhang, Bailey Miller, Kai Yan, Ioannis Gkioulekas, Shuang Zhao

ACM Transactions on Graphics (*SIGGRAPH 2020*), 39(4)

MULTI-SCALE APPEARANCE MODELING OF GRANULAR MATERIALS WITH CONTINUOUSLY VARYING GRAIN PROPERTIES

Cheng Zhang, Shuang Zhao

Eurographics Symposium on Rendering (*EGSR*), 2020

A DIFFERENTIAL THEORY OF RADIATIVE TRANSFER

Cheng Zhang, Lifan Wu, Changxi Zheng, Ioannis Gkioulekas, Ravi Ramamoorthi, Shuang Zhao

ACM Transactions on Graphics (*SIGGRAPH Asia 2019*), 38(6)

FontCode: Embedding Information in Text Documents Using Glyph Perturbation

Chang Xiao, **Cheng Zhang**, Changxi Zheng

ACM Transactions on Graphics (*SIGGRAPH 2018*), 37(2)

Patents

SYSTEMS AND METHODS FOR STEGANOGRAPHY BASED ON TEXT FONTS

Changxi Zheng, Chang Xiao, **Cheng Zhang**

US Patent 10,755,375

METHOD AND DEVICE FOR EFFICIENT BUILDING FOOTPRINT DETERMINATION

Lincan Zou, Liu Ren, Zeng Dai, **Cheng Zhang**

US Patent 10,553,025

Employment

Meta Reality Labs

RESEARCH SCIENTIST

Topic: Computer graphics research (e.g., differentiable rendering, inverse rendering), 3D reconstruction

Redmond, WA

July 2022 - Present

NVIDIA Corporation

APPLIED DEEP LEARNING RESEARCH (ADLR) GROUP: RESEARCH INTERN

Topic: Differentiable rendering on GPU

Remote

June 2021 - November 2021

Facebook Reality Labs

GRAPHICS GROUP: RESEARCH INTERN

Topic: 3D reconstruction, differentiable rendering

Redmond, WA

June 2019 - September 2019

Robert Bosch LLC

HMI GROUP: MOTION CAPTURE AND RECOGNITION INTERN

Topic: Automatic cycle detection and similarity evaluation of time series data

Sunnyvale, CA

June 2018 - September 2018

Robert Bosch LLC

HMI GROUP: GRAPHICS RESEARCH INTERN

Topic: Building footprints determination for real-time shadow generation

Palo Alto, CA

May 2017 - August 2017

Columbia University

COMPUTER GRAPHICS GROUP: GRADUATE RESEARCH ASSISTANT

Topic: Text editing, information embedding

New York, NY

June 2016 - April 2017

Reviewer

2026	ACM SIGGRAPH, Computers & Graphics, ACM SIGGRAPH Asia (Technical Papers Committee)
2025	ACM SIGGRAPH, ACM SIGGRAPH Asia
2024	ACM SIGGRAPH, ACM SIGGRAPH Asia, EGSR
2023	Nature Scientific Reports, ACM SIGGRAPH Asia, Pacific Graphics
2022	ACM SIGGRAPH
2021	ACM SIGGRAPH, Computer Graphics Forum
2020	ACM SIGGRAPH Asia

Honors & Awards

2023	ACM SIGGRAPH Outstanding Doctoral Dissertation Award	Worldwide
2021	Facebook Fellowship	Worldwide
2013	Beichuan Academic Scholarship	Beijing University of Technology
2011-2014	Academic Excellence Award	Beijing University of Technology

Skills

Programming	C, C++, Python, Matlab, T _E X, CUDA
Software and Applications	Mitsuba, PyTorch, Unity3D, Blender, OptiX